

Healthcare Data Cleaning, Transformation, Analysis & Visualization

1. Identified missing values
Checked the Medical Examinations and Hospitalization Details tables for missing values marked with “?”.
2. Filled missing Month and Year values
Filled missing Month values with Sep and missing Year values using the average year rounded to the nearest integer.
3. Filled missing categorical values
Identified the most frequently occurring values for smoker, Hospital tier and City tier and used them to fill missing entries.
4. Handled missing State ID
Checked missing State ID values and used “Unknown” where a State ID was not available.
5. Split Customer Names
Separated the names into Title, First Name and Last Name.
6. Converted NumberOfMajorSurgeries
Converted NumberOfMajorSurgeries into numerical values by replacing non-numeric entries with meaningful numerical values.
7. Checked inconsistent values
Checked the Heart Issues and smoker columns for inconsistent entries and standardized them where required.
8. Created Weight Status
Created Weight Status using BMI: Below 18.5 = Underweight; 18.5–24.9 = Normal Weight; 25.0–29.9 = Overweight; 30.0 and above = Obesity.
9. Created Diabetes Status
Created Diabetes Status using HBA1C: Below 5.7 = Normal; 5.7–6.4 = Prediabetes; 6.5 and above = Diabetes.
10. Created Date of Birth
Combined Year, Month and Date into one Date of Birth column and formatted it as DD-MMM-YYYY.
11. Calculated Age
Calculated customer age using Date of Birth and the dataset collection date, 8 June 2023.
12. Formatted healthcare charges
Formatted the charges column as currency in dollars (\$).
13. Created the Healthcare sheet
Created a new sheet named Healthcare and combined the three tables using Customer ID as the common field.
14. Used VLOOKUP
Used VLOOKUP with Customer ID and exact matching (FALSE) to bring the required fields from the three source tables into the Healthcare sheet.
15. Analysis – Cancer and Smoking
Created a PivotTable to examine cancer history among smokers and non-smokers and visualized it using a Pie/Donut chart.

16. Analysis – Transplants

Created a PivotTable comparing patients with and without transplants using total major surgeries and average HBA1C, then visualized the comparison.

17. Analysis – Weight and Diabetes

Created a PivotTable to compare healthcare charges across Weight Status and Diabetes Status and created a Column/Bar chart.

18. Analysis – Hospital Tier and State

Created a PivotTable to compare average healthcare charges for each Hospital tier within different State IDs and created a Column/Bar chart.

19. Analysis – Age and BMI

Created a Scatter Plot to study the relationship between Age and BMI. A linear trendline and R-squared value were added.

20. Analysis – Age and HBA1C

Created a Scatter Plot to study the relationship between Age and HBA1C. A linear trendline and R-squared value were added.

21. Analysis – Age and Charges

Created a Scatter Plot to study the relationship between Age and healthcare charges. A linear trendline and R-squared value were added.

22. Created Dashboard

Created a separate Dashboard sheet and arranged the key charts together for easy interpretation.

23. Added Slicers

Added slicers for Weight Status and Diabetes Status to allow interactive filtering.

24. Connected the Slicers

Connected the slicers to the relevant PivotTables so that selecting a Weight Status or Diabetes Status updates the connected visualizations.

25. Final checking and formatting

Checked formulas, PivotTables, charts and the dashboard for errors and ensured that headings, chart titles, currency values and layout were clear.

Main Excel tools used

- VLOOKUP – to combine information using Customer ID.
- PivotTable – to summarize and compare healthcare data.
- PivotChart – to create visualizations.
- Scatter Plot – to study relationships between Age and BMI, HBA1C and charges.
- Slicers – to filter the dashboard using Weight Status and Diabetes Status.
- Formatting and Find/Replace – to clean and present the data.